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Analysis of Hemodynamic Changes from Aneurysm Inception to Large Sizes

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Abstract

Background and Purpose—While previous studies have identified many risk factors for the progression and rupture of cerebral aneurysms, the changes in aneurysm flow characteristics during its evolution are not fully understood. This work analyzes the changes in the aneurysm hemodynamic environment from its initial development to later stages when the aneurysm has substantially enlarged.

Materials and Methods—A total of 88 aneurysms at four locations were studied with image based CFD. Two synthetic sequences representing the aneurysm geometry at three earlier stages were generated by shrinking the aneurysm sac while keeping the neck fixed or shrinking the neck simultaneously. The flow conditions were then quantitatively compared between these two modes of evolution.

Results—As aneurysms enlarged, the inflow rate increased in growing neck sequences, but decreased in fixed neck sequences. The inflow jet became more concentrated in both sequences. The mean aneurysm flow velocity and wall shear stress decreased in both sequences, but they decreased faster in enlarging aneurysms if the neck was fixed. Additionally, the intra-aneurysmal flows became more complex and more unstable, wall shear stress distribution became more oscillatory, and the area under low wall shear stress increased for both sequences.

Conclusion—The evolution of flow characteristics of aneurysms with fixed and growing necks are different. The observed trends suggest that fixed neck aneurysms may evolve towards a flow environment characteristic of stable aneurysms faster than aneurysms with growing necks, which could also evolve towards a more unfavorable environment.

Keywords

Cerebral Aneurysm; Aneurysm Growth; Hemodynamics; Evolution

Introduction

Intracranial aneurysm rupture and subsequent subarachnoid hemorrhage (SAH) constitutes a devastating event with high mortality and morbidity [1, 2]. Although the incidence of cerebral aneurysms is relatively high (cadaver studies suggest that approximately 2-3% of the population may harbor aneurysms [3, 4]), many aneurysms remain stable and never rupture (only about 1.3% of aneurysms rupture per year [5-7]). Therefore, it is very important to understand the conditions that predispose aneurysms for stabilization or destabilization, and subsequent rupture, so that the most appropriate management strategy (i.e. immediate treatment or conservative observation) is selected for each individual aneurysm.

Previous clinical studies have identified many risk factors such as aneurysm size, aneurysm location, presence of blebs, patient population, previous history of SAH, smoking, and hypertension [8-11]; and risk scales have been proposed [12]. Yet, this kind of studies have not significantly improved our knowledge of the underlying mechanisms responsible for the progression and ultimately rupture of cerebral aneurysms, which remain poorly understood. Furthermore, known risk factors or risk factor-based scoring scales are not sensitive and specific enough [13].

Aneurysm hemodynamics and shape is being investigated as possible factors for rupture [14-16]. Hemodynamics has been thought to play an important role in these mechanisms, and it has been proposed not only as a risk factor for rupture [17, 18], but as one of the main factors driving the growth and remodeling of the aneurysm wall [19-22]. However, little is known about how hemodynamic characteristics change as the aneurysm enlarges from its initial inception to later stages where it may stabilize or continue to grow, and eventually rupture. This is a difficult problem to study because very few aneurysms are followed longitudinally without treatment, and the ones that are followed are immediately intervened if any changes in their size or shape are observed. Furthermore, one very rarely has the chance of observing an aneurysm progression from its formation to a fully developed aneurysm resembling those that are detected incidentally (except perhaps in few de novo aneurysm cases).

Thus, the purpose of this study was to analyze the changes in the aneurysm hemodynamic conditions as the aneurysm enlarges from its initial inception to later stages when the aneurysm has become substantially larger ($>8\text{mm}$). For this purpose we conducted a “parametric” study in which, starting with a cerebral aneurysm model constructed from 3D medical images, we created a series of aneurysm geometries representing the aneurysm at earlier stages of their evolution (when the aneurysm was smaller), and analyzed the corresponding intra-aneurysmal flow characteristics using computational fluid dynamics (CFD).

When generating series of aneurysm geometries representing aneurysms at earlier stages of their evolution, one has the option of maintaining the aneurysm neck fixed or allowing it to enlarge along with the aneurysm. We believe that both evolution scenarios are plausible because some large aneurysms have small necks (see Fig. 1-a, b), which in all likelihood

did not evolve over time (or at least evolved very slowly in comparison to the aneurysm enlargement), while others (see Fig. 1-c, d) have quite large necks that most likely started small and grew as the aneurysm enlarged. Large aneurysms with intermediate sized necks could have evolved in any of the two manners. Therefore, we investigated the effect that an enlarging neck could have in the hemodynamic evolution of the aneurysm in comparison with an evolution where the neck does not change substantially as the aneurysm enlarges.

This study does not pretend to represent the exact evolution of a given aneurysm, but it includes the flow dynamics in patient-specific arterial geometries, as well as realistic aneurysm geometries at different locations where aneurysms actually developed. This allows for example to investigate whether the hemodynamic changes in aneurysms at different locations known to be associated with different rupture risk are similar or different.

Methods

Data

From our database of cerebral aneurysms imaged with 3D rotational angiography (3DRA) and modeled with image-based computational fluid dynamics [23], aneurysms larger than 8 mm were selected for this study at four different anatomical locations: anterior communicating artery (ACOM), posterior communicating artery (PCOM), basilar artery (BA) tip, and middle cerebral artery (MCA) bifurcation. In each location, 11 unruptured and 11 ruptured aneurysms were selected for a total of 88 aneurysms. Aneurysm and patient characteristics are summarized in Table 1.

Modeling

The patient-specific aneurysm models reconstructed from the 3DRA images of previous studies [23] were used to generate synthetic sequences representing the aneurysm at earlier stages when it was smaller. Specifically, for each aneurysm, three new geometries were generated by applying a Laplacian smoothing operator to the aneurysm surface, which shrinks the aneurysm while keeping the aneurysm neck (and the rest of the arterial model) fixed. A second sequence was generated by cutting off the aneurysm from the parent artery and applying Laplacian smoothing to this isolated aneurysm, which shrinks the aneurysm and its neck simultaneously, and then re-inserting the shrunk aneurysm back into the vascular model with a Boolean (merging) operator. In what follows, these two sequences are denoted “fixed neck” and “growing neck” sequences, respectively. They represent two possible geometric evolutions of the given aneurysm, one in which the aneurysm enlarges without any change in its neck, and another one in which the neck grows as the aneurysm enlarges. In both cases it was assumed that the geometry of the parent artery does not change during the growth of the aneurysm. To illustrate the methodology, an example of both scenarios is presented in Fig. 2 for an aneurysm in the MCA.

Subsequently, unstructured grids composed of tetrahedral elements with a minimum resolution of 0.2 mm were automatically generated for each stage of the two sequences of each aneurysm using an advancing front method implemented in an in-house software [24]. Then, CFD simulations were carried out for the 4 stages of each of the two sequences

for each aneurysm (a total of 616 simulations since the last stage of both sequences is the same). In each simulation, the 3D unsteady incompressible Navier-Stokes equations were numerically solved using a fully implicit finite element solver implemented in an in-house software. Blood was modeled as an incompressible Newtonian fluid with density $\rho = 1.0 \text{ g/cm}^3$ and viscosity $\mu = 0.04 \text{ Poise}$. Arterial and aneurysm walls were approximated as rigid. Pulsatile flow waveforms obtained from phase-contrast MR measurements in healthy subjects [25] were scaled based on an empirical power law relating mean flow rate and vessel cross-sectional area [26], and prescribed as inflow boundary conditions using the Womersley profile [27] at the inlet located either at the internal carotid artery (ICA) for aneurysms in the anterior circulation, or at the vertebral artery (VA) for posterior circulation aneurysms. Outlet boundary conditions were set as pressure and outflow rates to specify flow divisions consistent with Murray's law. No-slip boundary conditions were prescribed at the vascular walls.

For each aneurysm, the same boundary conditions were used for all stages of each evolutionary sequence. All simulations were performed for two cardiac cycles with a heart rate of 60 beats per minute and a timestep size of 0.01 sec. The results from the second cycle were subsequently used for hemodynamic characterization and comparison. For further methodological details and discussion of the main assumptions and approximations, see [28].

Analysis

In order to characterize and compare the hemodynamic environment at each stage of the evolution of each aneurysm, different hemodynamic variables were computed from the results of the CFD simulations. These variables were used to characterize the inflow jet into the aneurysm (strength and concentration), the intra-aneurysmal flow pattern (flow speed, complexity, and instability), and the wall shear stress distribution (intensity, uniformity, concentration, and oscillation). A list of the hemodynamic variables used in this study is presented in Table 2. The exact mathematical definitions of these variables and detailed descriptions of the procedure to calculate them can be found in [29, 30].

Once these variables were computed for each stage of an evolutionary sequence of a given aneurysm, their "rate of change" were calculated by fitting a simple linear regression model for each variable and for each sequence. The mean "rate of change" was calculated by averaging the slopes of the fitted lines for aneurysms at different locations, for both growing and fixed neck sequences. Since the time scale between different stages of aneurysm evolution in these synthetic sequences is not known, the term "rate of change" should be interpreted as relative change with respect to aneurysm size rather than time. Similarly, the terms "speed", "faster", and "slower" indicate rate of change with respect to aneurysm size not time. Fig. 3 shows, as an example, the evolution of mean WSS in ACOM aneurysms for both the fixed neck (left) and growing neck (right) sequences, along with the mean "rate of change" (dashed lines) for unruptured (green) and ruptured (red) aneurysms. In this example, the average "speed" of WSS decrease is "faster" in aneurysms with fixed necks (left panel) compared to aneurysms with growing necks (right panel).

Subsequently, statistical analysis was performed to explore differences in the mean “rate of change” of hemodynamic variables between aneurysm with growing or fixed necks, in order to investigate whether the flow conditions in aneurysms with growing necks and aneurysms with fixed necks would evolve differently as the aneurysms enlarge. For this purpose, a paired t-test was performed to determine whether differences in the mean “rate of change” of hemodynamic variables of the two sequences were statistically significant ($p < 0.05$).

Since it is known that aneurysms at the anterior communicating artery carry significantly higher risk of rupture than aneurysms at the middle cerebral artery bifurcation [31], t-tests were also performed to compare the mean “rate of change” of hemodynamic variables between ACOM and MCA aneurysms, in order to determine whether their hemodynamic evolution could be different. All statistical analysis was carried out using R scripts.

Results

Comparisons of the “rate of change” of hemodynamic characteristics during aneurysm enlargement between fixed neck and growing neck scenarios, for different aneurysm locations are presented in Tables 3-6 and summarized in Fig. 4. Similarly, comparisons between ACOM and MCA aneurysms are presented in Tables 7-8 and summarized in Fig. 5.

To illustrate the differences between fixed and growing neck scenarios, an example is presented in Fig. 6 and 7. Figure 6 shows the evolution of the WSS magnitude as the aneurysm enlarges, for both the fixed neck (a) and the growing neck (b) scenarios; while Fig. 7 shows the corresponding flow patterns. It can be clearly seen that the hemodynamic evolution is quite different if the neck enlarges or not.

Evolution of Inflow Jet Characteristics

The mean inflow rate into the aneurysm (Q) increased for growing neck sequences but decreased for fixed neck sequences at all locations. The rate of increase in growing neck sequences was “faster” than the rate of decrease in fixed neck sequences in ACOM ($p < 0.001$), PCOM ($p < 0.001$), and BA ($p < 0.001$) but was “slower” in MCA ($p < 0.001$) aneurysms. Compared to MCA aneurysms, the inflow rate increased “faster” in ACOM aneurysms with enlarging necks ($p = 0.005$); however, the difference in the “speeds of change” of the inflow rate between ACOM and MCA aneurysms with fixed necks did not reach statistical significance. The inflow jet became more concentrated (larger ICI) as the aneurysm enlarged at all locations, but it became more concentrated “faster” in growing neck sequences than in fixed neck sequences at ACOM ($p < 0.001$), PCOM ($p < 0.001$), and MCA ($p < 0.001$) while it became more concentrated more “slowly” at the BA ($p < 0.001$).

Evolution of Intra-Aneurysmal Flow Characteristics

As the aneurysms enlarged, the intra-aneurysmal flows became more complex (larger corelen) and more unstable (larger podent) for both growing and fixed neck sequences at all locations; but there was no statistically significant differences between the “rates” of increase of these variables between growing and fixed necks sequences or between ACOM and MCA aneurysms.

On the other hand, the mean aneurysm flow velocity (V_{mean}) decreased in both growing and fixed neck sequences at all locations, but decreased “faster” for fixed neck sequences (ACOM: $p=0.003$, PCOM: $p<0.001$, BA: $p<0.001$, and MCA: $p=0.05$). In contrast, the maximum flow velocity (V_{max}) increased in growing neck sequences at all locations; it also increased in fixed neck sequences at the ACOM and BA but decreased in fixed neck sequences at the PCOM and MCA. The differences in the “speeds of change” of the maximum velocity between growing and fixed neck sequences was statistically significant in ACOM: $p=0.01$, PCOM: $p<0.001$, and BA: $p=0.02$, but these differences did not reach statistical significance in MCA or between ACOM and MCA aneurysms.

Evolution of Wall Shear Stress Characteristics

The mean wall shear stress (WSS) decreased in both growing and fixed neck sequences at all locations, but it decreased “faster” in fixed neck sequences in ACOM: $p=0.009$, PCOM: $p<0.001$, and BA: $p=0.008$. However, the difference in the “speeds of change” of the mean wall shear stress between growing and fixed neck sequences did not reach statistical significance in MCA. WSS also tended to decrease “faster” in MCA compared to ACOM aneurysms for both growing and fixed neck sequences, but these differences did not reach statistical significance.

As the aneurysms enlarge, the wall shear stress distribution becomes more oscillatory (larger OSI) in both growing and fixed neck sequences at all locations. The “speed” at which the WSS becomes more oscillatory were not significantly different between growing and fixed neck sequences, except at the PCOM where OSI increased “faster” in fixed neck sequences ($p=0.04$). Additionally, OSI tended to increase “faster” in ACOM compared to MCA aneurysms in fixed neck sequences but this difference did not reach statistical significance.

The area under low wall shear stress (LSA) increased in both growing neck and fixed neck sequences at all locations. No significant differences in the “speed” of increase of LSA were found between growing and fixed neck sequences, except at the PCOM, where LSA increased “faster” in fixed neck sequences. In addition, LSA tended to increase “faster” in ACOM compared to MCA aneurysms for both growing and fixed neck sequences. In addition, as the aneurysm enlarged with a fixed neck the WSS became more concentrated (SCI increased), irrespective of location. However, if the neck enlarged, the WSS also became more concentrated (SCI increased) in ACOM and PCOM aneurysms but became more diffuse (SCI decreased) in BA and MCA aneurysms.

Discussion

As mentioned before, hemodynamics is thought to play an important role in the evolution and eventual rupture of cerebral aneurysms. However, it is largely unknown how hemodynamic conditions change as the aneurysm grows from its initial inception to reach large sizes. In this study we analyzed these changes using image based CFD and synthetic sequences to represent the enlargement of cerebral aneurysms at different locations, and in particular investigated the effects of neck enlargement during the aneurysm progression.

In a previous study [32], two different growth types were observed for 6 small cerebral aneurysms during follow-up observations; focal aneurysm growth with bleb or blister formation, and global aneurysm growth accompanied by neck enlargement, and different hemodynamic conditions were observed for each growth type. In our study, we considered two possible evolution scenarios for each aneurysm to investigate the effects of neck enlargement during the aneurysm evolution. While we analyzed the changes in the aneurysm hemodynamic conditions when the aneurysm enlarges considerably, compared to the small enlargement observed in longitudinal image data in the previous study, we agree that the evolution of flow characteristics of aneurysms with fixed and growing necks are different.

Our results indicate that in general, as the aneurysm enlarges, the inflow jet becomes more concentrated, the intra-saccular flow pattern becomes more complex and unstable, average aneurysm flow velocity becomes slower, and the wall shear stress becomes lower in average but more oscillatory and with larger areas under low WSS, irrespective of aneurysm location or neck evolution. However, if the aneurysm neck grows as the aneurysm enlarges, the inflow jet becomes stronger and the maximum aneurysm flow velocity increases, the inflow jet becomes more concentrated “faster”, and the average velocity and wall shear stress tend to decrease more “slowly” compared to aneurysms with fixed necks, irrespective of aneurysm location. Furthermore, in ACOM aneurysms with growing necks the inflow jet becomes stronger “faster” than in MCA aneurysms with growing necks. For aneurysms with fixed necks, although the inflow rate decreased as the aneurysm enlarged, the maximum aneurysm flow velocity increased in ACOM aneurysms while it decreased in MCA aneurysms. The average WSS decreased and OSI increased in both MCA and ACOM aneurysms for both fixed and growing neck sequences. For aneurysms with growing necks, the wall shear stress concentration increased in ACOM aneurysms but decreased in MCA aneurysms.

These findings suggest that, in general, aneurysms with growing necks progress towards flow conditions with larger and more concentrated inflow jets, higher maximum velocity, than aneurysms with fixed necks; and that these changes tend to occur at a “faster” rate. In contrast, although, the mean velocity and WSS decrease as the aneurysm enlarges, these changes seem to be “slower” if the neck also enlarges. The results also suggest that ACOM aneurysms are exposed to larger and more concentrated inflow jets, higher velocity and WSS, for “longer periods” (i.e. for more intermediate size stages) than MCA aneurysms, or conversely, that MCA aneurysms progress to hemodynamic environments with lower and more diffuse inflow jets, lower velocity and WSS “faster” than ACOM aneurysms. This seems to indicate that ACOM aneurysms, especially those with enlarging necks, may evolve towards and be subjected to more adverse hemodynamic environments (i.e. larger and more concentrated inflow jets, higher and more heterogeneous WSS distributions, and higher speed flow) for “longer periods” than MCA aneurysms. Conversely, MCA aneurysms seem to evolve towards a less hostile hemodynamic environment (i.e. more diffuse and weaker inflow jets and more uniform WSS distributions), especially those with fixed necks, “faster” than ACOM aneurysms that are exposed to more hostile conditions for “longer periods” during their enlargement. This could explain the overall increased relative rupture risk of ACOM aneurysms compared to MCA aneurysms.

Interestingly, it was noticed that in a few cases, the mean WSS initially decreased but then increased as the aneurysm enlarged with a growing neck, but not in the corresponding fixed neck sequence. One such example can be seen in Fig. 3 (right panel). To understand this behavior, the inflow jet and WSS distributions at different stages of the evolution are presented in Fig. 8, for both scenarios. It can be seen that when the neck enlarges (c, d), the inflow jet is remarkably affected as well as the associated flow impingement region and the WSS distribution. In particular, the flow impingement shifts from the proximal part of the aneurysm body at an early stage (c, d) to the distal part of the dome at a larger stage (e, f). However, with a fixed large neck (a, b), the inflow jet is not so dramatically affected throughout the evolution of the aneurysm. Once again, this illustrates that the evolution of the hemodynamic environment of aneurysms with growing and fixed necks can be substantially different.

We also investigated the differences in average “rate of change” of hemodynamic variables between ruptured and unruptured aneurysms for both fixed neck and growing neck sequences at each location. Although we observed some differences in the “speed” of change of hemodynamic variables for ruptured and unruptured aneurysms, the small sample size prevents us from drawing any definite conclusion. These differences should be further investigated with more data in each location.

Our study has several limitations. The CFD analysis makes a number of typical simplifying approximations such as Newtonian fluid, non-compliant walls, and non-patient specific inflow and outflow boundary conditions. Additionally, in this study the geometric evolution of the aneurysm was approximated by synthetically creating aneurysm geometries to represent earlier stages of the aneurysm. While this approach provides information regarding the general trends of the evolution of hemodynamic characteristics, it does not intend to represent the exact evolution of each aneurysm. Furthermore, in these sequences, the temporal profiles of hemodynamic variables exhibited variability, but to simplify the analysis and compare the overall tendency of hemodynamic characteristics, a simple linear regression was used to estimate the change in these quantities as the aneurysm enlarges. Additionally, the time scales between different stages of aneurysm evolution in these synthetic sequences is not known, therefore, the terms “speed” and “rate of change” should be interpreted in terms of relative enlargement or size rather than time.

Conclusion

The evolution of flow characteristics of aneurysms with fixed and growing necks were different at all locations. As the aneurysms enlarged, the mean flow velocity and wall shear stress decreased, and this decrease occurred “faster” in aneurysms with fixed necks. The flow complexity, instability, and oscillation increased, but there was no significant difference in the “speed” of increase between aneurysms with growing and fixed necks. Additionally, the inflow rate increased in aneurysms with growing necks and the inflow jet became more concentrated as the aneurysm enlarged. These observations suggest that aneurysms with fixed necks may evolve towards a flow environment characteristic of stable aneurysms “faster” than aneurysms with growing necks, which could also evolve towards

a more unfavorable environment, thus perhaps placing aneurysms with growing necks at a higher risk of destabilization and rupture.

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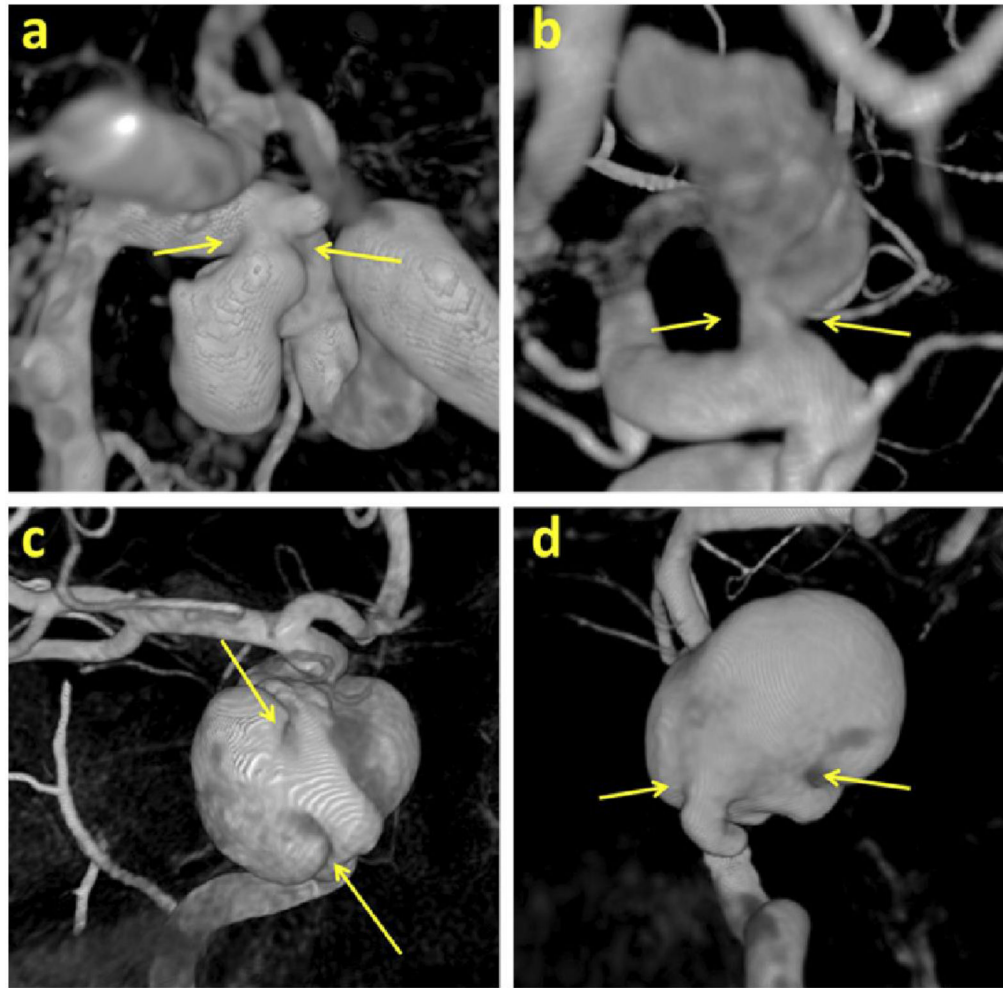


Figure 1:
Examples of large aneurysms with small necks where it is likely that the neck is fixed (a, b), and large aneurysms with large necks where it is likely that the neck enlarges along with the aneurysm (c, d). Visualizations are volume rendered 3D rotational angiographies. Arrows point to the aneurysm neck.

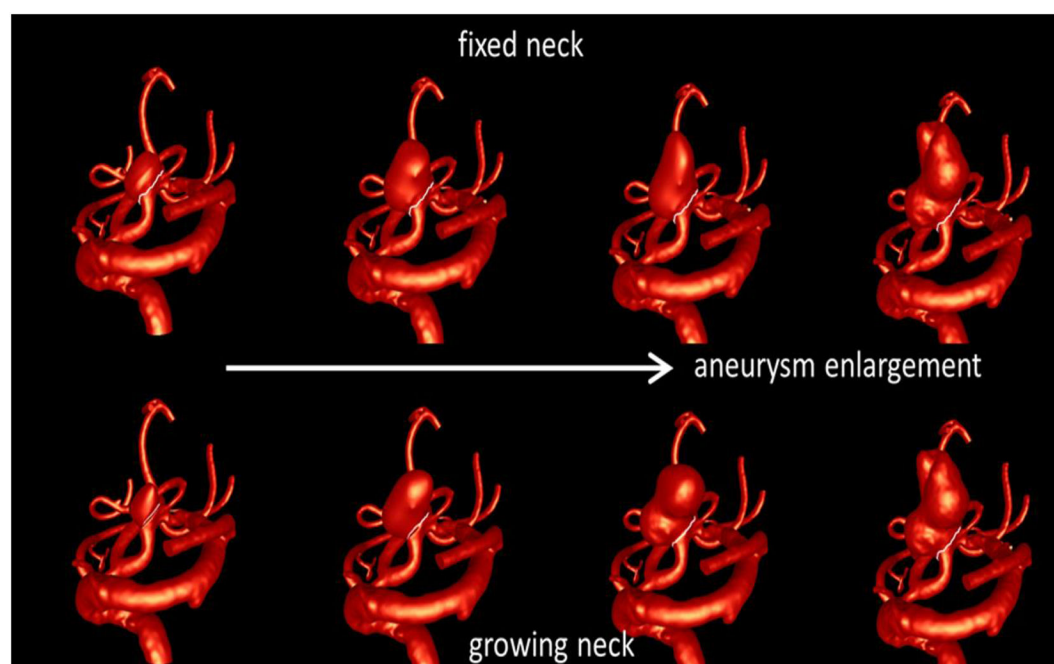


Figure 2.
Example of evolutionary sequences or scenarios with fixed (top) and growing (bottom) neck for an aneurysm in the MCA.

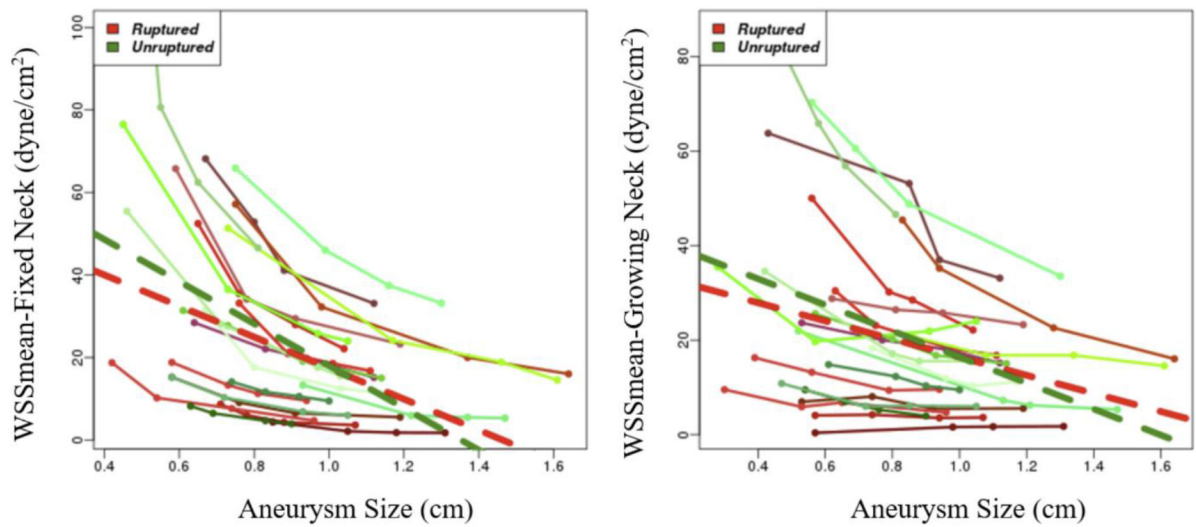
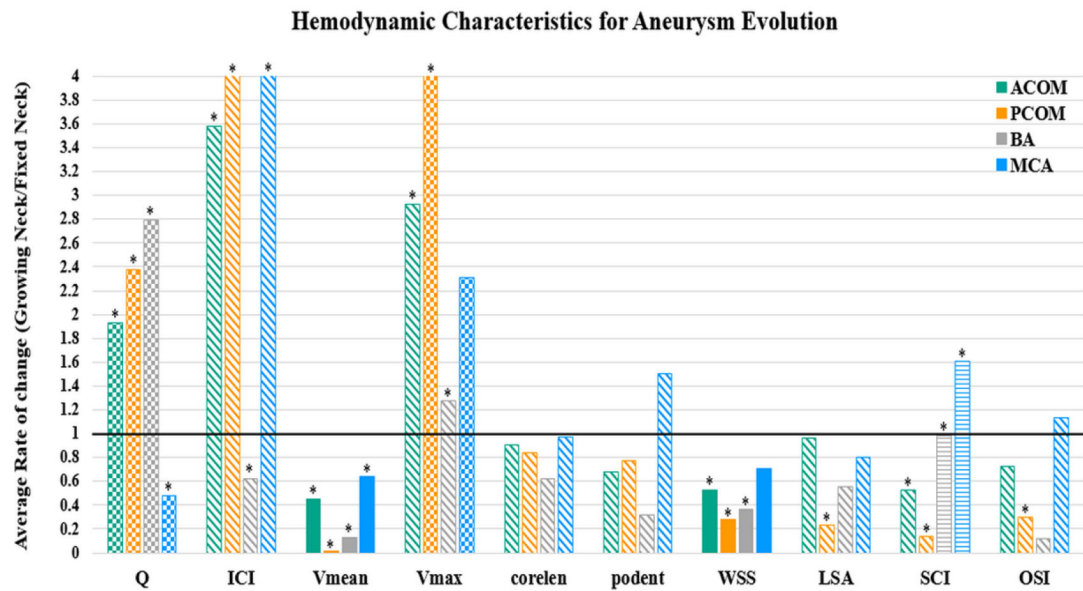


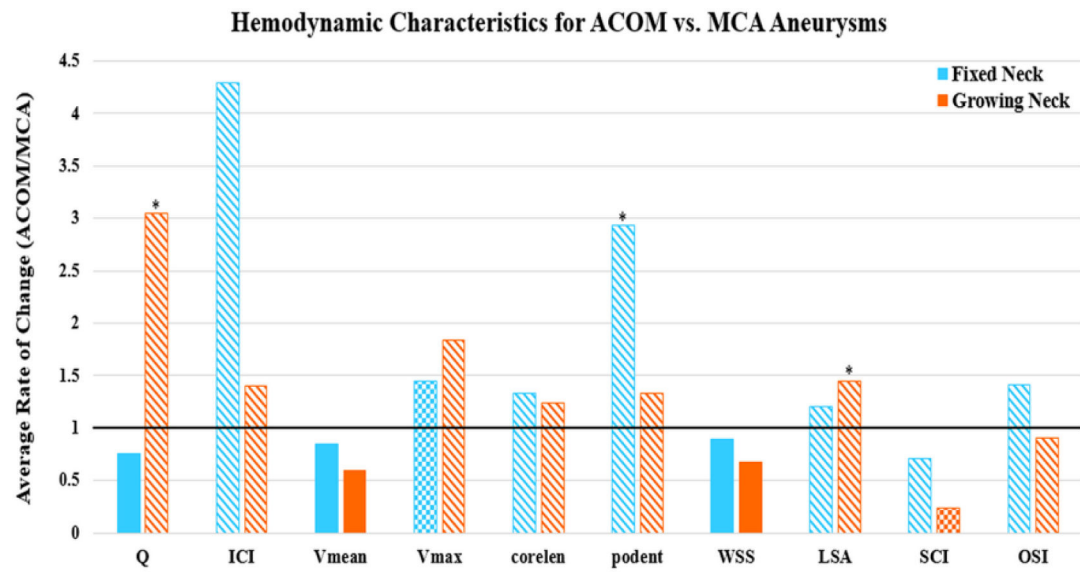
Figure 3:
Change of mean WSS in ruptured (red) and unruptured (green) ACOM aneurysms with fixed and growing necks (solid lines), and corresponding average “rate of change” (dashed lines).



- Bars with diagonal patterns show positive average “rate of change” in both growing and fixed neck cases.
- Bars with solid fills present negative average “rate of change” in both fixed and growing neck cases.
- Checked bars illustrate positive average “rate of change” in growing neck and negative average “rate of change” in fixed neck cases.
- Bars with horizontal lines present negative average “rate of change” in growing neck and positive average “rate of change” in fixed neck cases.

Figure 4:

Average “rate of change” of hemodynamic variables in four locations; ACOM, PCOM, BA, and MCA. Statistically significant differences ($P < .05$) are indicated with a “*”.



- Bars with diagonal patterns show positive average “rate of change” in both ACOM and MCA aneurysms.
- Bars with solid fills present negative average “rate of change” in both ACOM and MCA aneurysms.
- Checked bars illustrate positive average “rate of change” in ACOM and negative average “rate of change” in MCA aneurysms.

Figure 5:

Average “rate of change” of hemodynamic variables in ACOM and MCA. Statistically significant differences ($P < .05$) are indicated with a “*”.

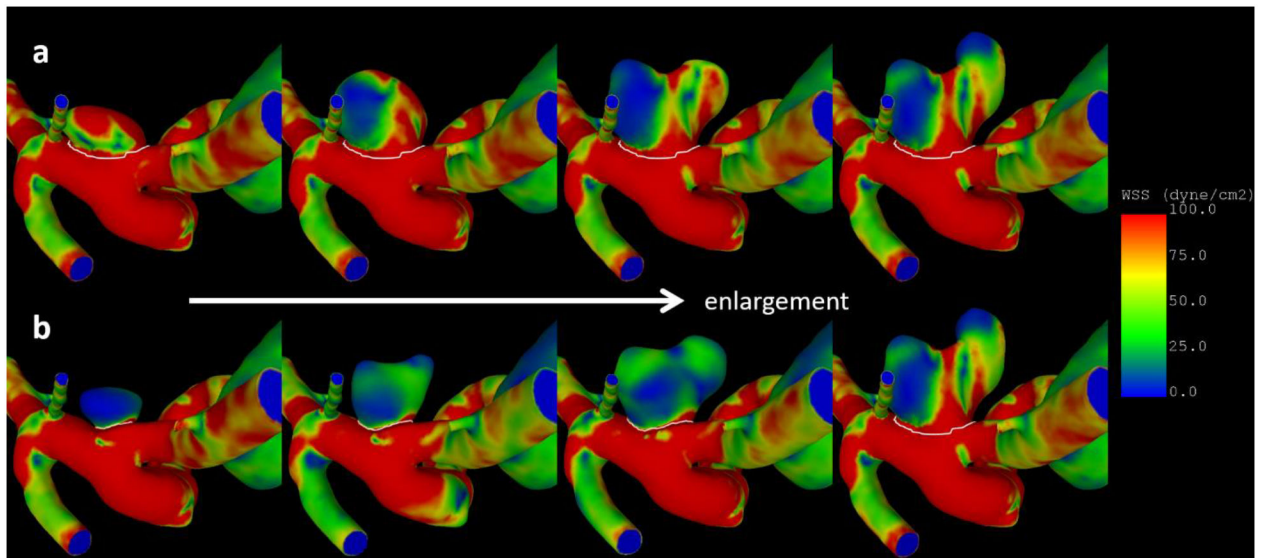


Figure 6: WSS distribution during an aneurysm enlargement when the neck is fixed (a) and when the neck enlarges together with the aneurysm (b).

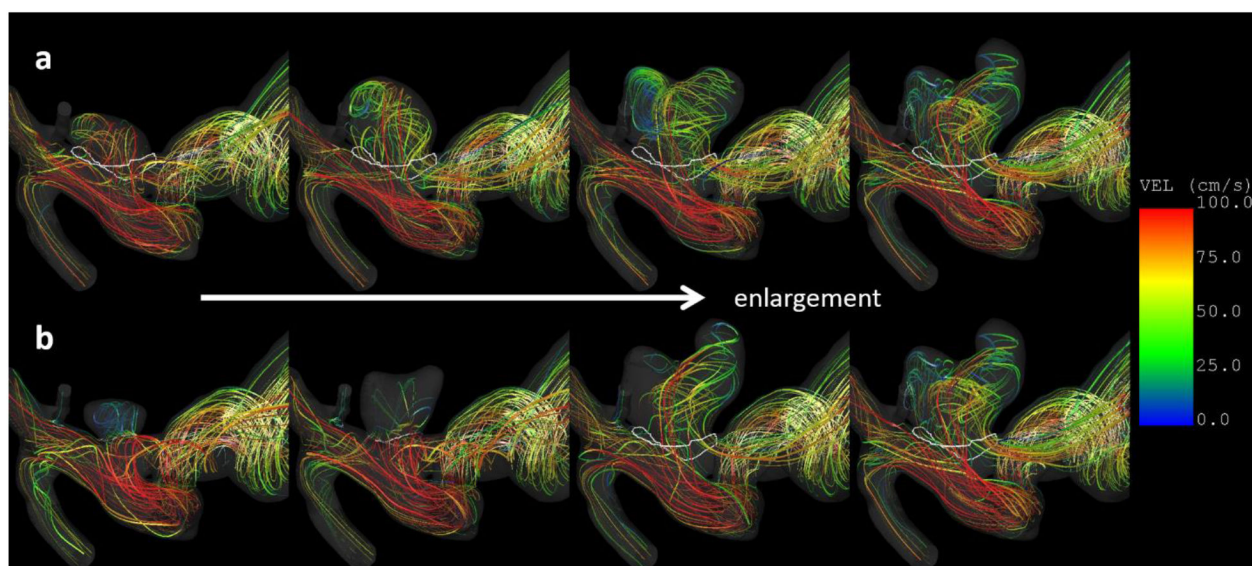


Figure 7:
Flow structure during an aneurysm enlargement when the neck is fixed (a) and when the neck enlarges together with the aneurysm (b).

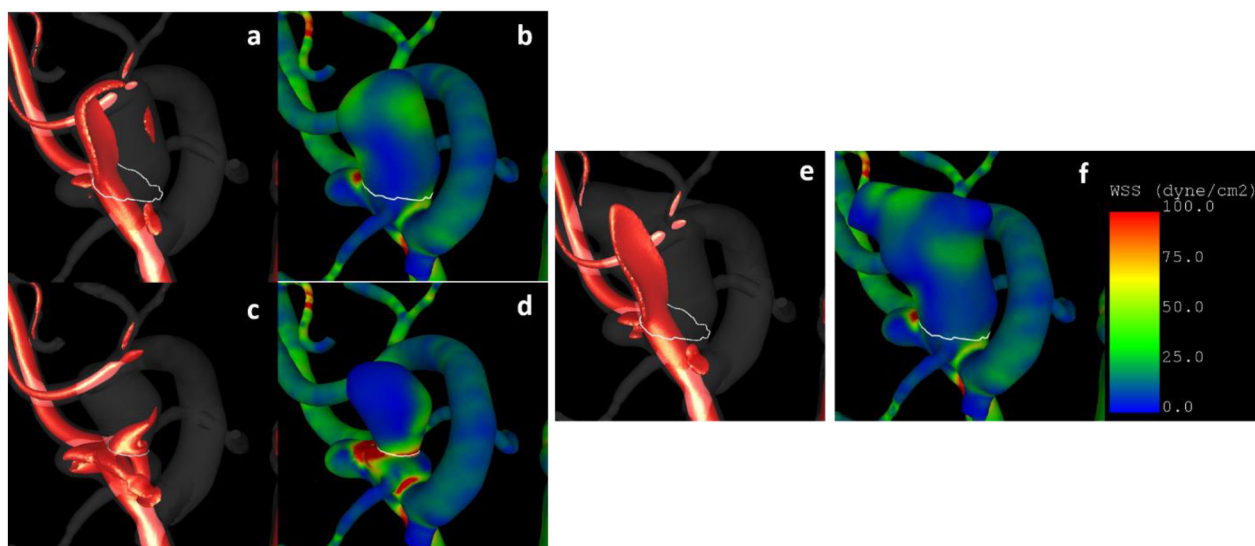


Figure 8.

Example showing effect of neck enlargement on inflow jet and WSS distribution. Inflow jet at early stage with fixed neck (a), growing neck (c), and at later stage (e). Corresponding WSS distributions at early stage with fixed neck (b), growing neck (d), and at later stage (F)

Table 1:

Summary of patients and cerebral aneurysm characteristics. *ACOM* anterior communicating artery, *PCOM* posterior communicating artery, *MCA* middle cerebral artery, *BA* basilar artery, *AR* aspect ratio. The mean $\pm$ SD is reported for the known age.

Variables	R	U
<i>Patient Variables</i>		
- Female (NO.) (%)	28 (31.8%)	27 (30.7%)
- Male (NO.) (%)	8 (9.1%)	7 (7.9%)
- Not known (NO.) (%)	8 (9.1%)	10 (11.4%)
*Age (yr.) (mean $\pm$ SD)	58 $\pm$ 12	67 $\pm$ 9.41
<i>Aneurysm Variables</i>		
Multiplicity (NO.) (%)	11 (12.5%)	10 (11.4%)
Location		
- ACOM (NO.) (%)	11 (12.5%)	11 (12.5%)
- PCOM (NO.) (%)	11 (12.5%)	11 (12.5%)
- MCA (NO.) (%)	11 (12.5%)	11 (12.5%)
- BA (NO.) (%)	11 (12.5%)	11 (12.5%)
Type		
- Sidewall (NO.) (%)	0	1 (1.1%)
- Bifurcation (NO.) (%)	44 (50%)	43 (48.9%)
Aneurysm Size (mm) (mean $\pm$ SD)	12.2 $\pm$ 3.1	12.8 $\pm$ 4.5
Neck Size (mm) (mean $\pm$ SD)	5.4 $\pm$ 1.6	6.4 $\pm$ 2.2
AR (mean $\pm$ SD)	1.86 $\pm$ 0.63	1.72 $\pm$ 0.58

Table 2:

Hemodynamic variables. The units are presented in the parentheses. The exact mathematical definitions of these variables and detailed descriptions of the procedures to compute them can be found in [29], details about corelen and podent can be found in [30].

Characterization	Variable	Definition
Inflow Jet	Q (cm ³ /s)	Mean inflow rate into aneurysm
	ICI	Inflow concentration index
Aneurysmal Flow Pattern	Vmean (cm/s)	Mean velocity
	Vmax (cm/s)	Peak velocity
	Corelen (cm)	Vortex core line length (flow complexity)
	Podent	Proper orthogonal decomposition (POD) entropy (flow stability)
Wall Shear Stress Distribution	WSS (dyne/cm ²)	Mean wall shear stress
	LSA (%)	Low shear area
	SCI	Shear concentration index
	OSI	Mean oscillatory shear index

Table 3:

Statistics of average rate of change of hemodynamic variables – ACOM location. Data are average rate of change (mean $\pm$ SD). *CI*, confidence interval. * Statistically significant difference (95% confidence).

Variable	Growing neck (mean $\pm$ SD)	Fixed neck (mean $\pm$ SD)	Difference in the Means [95 % CI]	P-value
Q	0.61 $\pm$ 0.28	-0.31 $\pm$ 0.28	-0.90 [-1.10, -0.73]	<0.001*
ICI	1.14 $\pm$ 0.56	0.32 $\pm$ 0.46	-0.84 [-1.16, -0.50]	<0.001*
Vmean	-5.88 $\pm$ 6.84	-13.04 $\pm$ 9.20	-6.33 [-10.33, -2.38]	0.003*
Vmax	13.36 $\pm$ 14.85	4.57 $\pm$ 10.36	-6.25 [-14.27, -1.25]	0.01*
corelen	5.23 $\pm$ 2.65	5.77 $\pm$ 3.39	0.33 [-1.39, 2.54]	0.86
podent	0.30 $\pm$ 0.26	0.44 $\pm$ 0.32	0.12 [-0.03, 0.28]	0.13
WSS	-23.37 $\pm$ 24.51	-44.31 $\pm$ 36.53	-16.05 [-30.89, -4.28]	0.009*
LSA	62.52 $\pm$ 34.54	64.64 $\pm$ 37.46	2.51 [-18.56, 25.61]	0.79
SCI	3.37 $\pm$ 5.55	6.42 $\pm$ 7.81	2.67 [0.60, 5.82]	0.01*
OSI	0.02 $\pm$ 0.03	0.03 $\pm$ 0.02	0.007 [-0.01, 0.03]	0.40

Table 4:

Statistics of average rate of change of hemodynamic variables – PCOM location. Data are average rate of change (mean $\pm$ SD). *CI*, confidence interval. * Statistically significant difference (95% confidence).

Variable	Growing neck (mean $\pm$ SD)	Fixed neck (mean $\pm$ SD)	Difference in the Means [95 % CI]	P-value
Q	0.78 $\pm$ 0.77	-0.33 $\pm$ 0.47	-0.92 [-1.35, -0.62]	<0.001*
ICI	0.76 $\pm$ 0.45	0.04 $\pm$ 0.29	-0.77 [-0.99, -0.50]	<0.001*
Vmean	-0.19 $\pm$ 6.02	-15.52 $\pm$ 10.54	-13.35 [-18.10, -9.02]	<0.001*
Vmax	12.03 $\pm$ 18.81	-0.48 $\pm$ 4.71	-9.20 [-14.37, -5.21]	<0.001*
corelen	4.82 $\pm$ 4.74	5.77 $\pm$ 5.66	0.88 [-1.66, 3.18]	0.44
podent	0.19 $\pm$ 0.30	0.25 $\pm$ 0.34	0.04 [-0.10, 0.19]	0.60
WSS	-10.31 $\pm$ 20.49	-37.34 $\pm$ 33.38	-21.16 [-31.87, -11.48]	<0.001*
LSA	13.91 $\pm$ 28.63	60.81 $\pm$ 37.71	47.62 [25.32, 70.89]	<0.001*
SCI	1.77 $\pm$ 10.92	12.72 $\pm$ 10.59	8.87 [4.38, 14.59]	<0.001*
OSI	0.005 $\pm$ 0.02	0.02 $\pm$ 0.02	0.01 [0.0003, 0.02]	0.04*

Table 5:

Statistics of average rate of change of hemodynamic variables – BA location. Data are average rate of change (mean $\pm$ SD). *CI*, confidence interval. * Statistically significant difference (95% confidence).

Variable	Growing neck (mean $\pm$ SD)	Fixed neck (mean $\pm$ SD)	Difference in the Means [95 % CI]	P-value
Q	1.04 $\pm$ 0.90	-0.37 $\pm$ 0.70	-1.16 [-1.66, -0.83]	<0.001*
ICI	2.57 $\pm$ 2.64	4.11 $\pm$ 17.36	-1.49 [-2.23, -0.84]	<0.001*
Vmean	-2.47 $\pm$ 7.51	-19.40 $\pm$ 25.74	-11.40 [-16.18, -6.76]	<0.001*
Vmax	16.18 $\pm$ 21.33	12.68 $\pm$ 46.12	-9.46 [-19.70, -1.27]	0.02*
corelen	5.08 $\pm$ 3.72	8.16 $\pm$ 15.26	0.21 [-2.48, 3.64]	0.86
podent	0.16 $\pm$ 0.35	0.49 $\pm$ 1.56	0.08 [-0.09, 0.21]	0.35
WSS	-20.21 $\pm$ 20.16	-55.71 $\pm$ 91.69	-17.73 [-32, -4.97]	0.008*
LSA	36.30 $\pm$ 32.97	65.34 $\pm$ 59.50	25.86 [-2.30, 63.47]	0.13
SCI	-10.51 $\pm$ 31.58	10.66 $\pm$ 39.75	6.44 [2.71, 11.04]	0.001*
OSI	0.004 $\pm$ 0.03	0.04 $\pm$ 0.15	0.008 [-0.006, 0.02]	0.22

Table 6:

Statistics of average rate of change of hemodynamic variables – MCA location. Data are average rate of change (mean $\pm$ SD). *CI*, confidence interval. * Statistically significant difference (95% confidence).

Variable	Growing neck (mean $\pm$ SD)	Fixed neck (mean $\pm$ SD)	Difference in the Means [95 % CI]	P-value
Q	0.20 $\pm$ 0.64	-0.41 $\pm$ 0.60	-0.54 [-0.90, -0.28]	<0.001*
ICI	0.82 $\pm$ 0.72	0.07 $\pm$ 0.88	-0.57 [-0.94, -0.27]	<0.001*
Vmean	-9.82 $\pm$ 9.17	-15.42 $\pm$ 10.43	-5.59 [-10.56, 0.02]	0.05*
Vmax	7.27 $\pm$ 23.84	-3.15 $\pm$ 24.52	-5.03 [-13.22, 2.18]	0.16
corelen	4.21 $\pm$ 3.64	4.33 $\pm$ 4.45	0.21 [-2.29, 2.35]	0.95
podent	0.23 $\pm$ 0.29	0.15 $\pm$ 0.39	0.007 [-0.19, 0.14]	0.97
WSS	-34.82 $\pm$ 34.96	-49.46 $\pm$ 39.52	-15.10 [-32.66, 2.77]	0.09
LSA	43.12 $\pm$ 39.08	53.60 $\pm$ 38.30	14.44 [-8.95, 39.42]	0.23
SCI	-14.43 $\pm$ 79.96	8.98 $\pm$ 9.14	5.16 [0.91, 12.66]	0.03*
OSI	0.03 $\pm$ 0.03	0.02 $\pm$ 0.04	-0.006 [-0.03, 0.01]	0.70

Table 7:

Statistics of average rate of change of hemodynamic variables – ACOM and MCA locations (Fixed Neck Sequences). Data are average rate of change (mean $\pm$ SD). *CI*, confidence interval. * Statistically significant difference (95% confidence).

Variable	ACOM (mean $\pm$ SD)	MCA (mean $\pm$ SD)	Difference in the Means [95 % CI]	P-value
Q	-0.31 $\pm$ 0.28	-0.41 $\pm$ 0.60	0.01 [-0.17, 0.21]	0.94
ICI	0.32 $\pm$ 0.46	0.07 $\pm$ 0.88	0.06 [-0.18, 0.32]	0.68
Vmean	-13.04 $\pm$ 9.20	-15.42 $\pm$ 10.43	2.12 [-2.37, 6.91]	0.41
Vmax	4.57 $\pm$ 10.36	-3.15 $\pm$ 24.52	2.68 [-2.16, 9.63]	0.27
corelen	5.77 $\pm$ 3.39	4.33 $\pm$ 4.45	1.38 [-0.83, 4.02]	0.23
podent	0.44 $\pm$ 0.32	0.15 $\pm$ 0.39	0.23 [0.07, 0.43]	0.006*
WSS	-44.31 $\pm$ 36.53	-49.46 $\pm$ 39.52	[-12.26, 21.79]	0.54
LSA	64.64 $\pm$ 37.46	53.60 $\pm$ 38.30	11.58 [-12.39, 36.05]	0.40
SCI	6.42 $\pm$ 7.81	8.98 $\pm$ 9.14	-0.46 [-4.36, 2.75]	0.74
OSI	0.03 $\pm$ 0.02	0.02 $\pm$ 0.04	0.01 [-0.003, 0.03]	0.11

Table 8:

Statistics of average rate of change of hemodynamic variables – ACOM and MCA locations (Growing Neck Sequences). Data are average rate of change (mean $\pm$ SD). *CI*, confidence interval. * Statistically significant difference (95% confidence).

Variable	ACOM (mean $\pm$ SD)	MCA (mean $\pm$ SD)	Difference in the Means [95 % CI]	P-value
Q	0.61 $\pm$ 0.28	0.20 $\pm$ 0.64	0.38 [0.12, 0.64]	0.005*
ICI	1.14 $\pm$ 0.56	0.82 $\pm$ 0.72	0.32 [-0.04, 0.72]	0.10
Vmean	-5.88 $\pm$ 6.84	-9.82 $\pm$ 9.17	2.85 [-1.58, 8.37]	0.19
Vmax	13.36 $\pm$ 14.85	7.27 $\pm$ 23.84	5.51 [-2.58, 15.37]	0.17
corelen	5.23 $\pm$ 2.65	4.21 $\pm$ 3.64	1.27 [-0.56, 2.88]	0.19
podent	0.30 $\pm$ 0.26	0.23 $\pm$ 0.29	0.08 [-0.06, 0.23]	0.24
WSS	-23.37 $\pm$ 24.51	-34.82 $\pm$ 34.96	6.03 [-4.75, 21]	0.28
LSA	62.52 $\pm$ 34.54	43.12 $\pm$ 39.08	24.56 [1.56, 46.29]	0.04*
SCI	3.37 $\pm$ 5.55	-14.43 $\pm$ 79.96	1.19 [-2.81, 4.17]	0.54
OSI	0.02 $\pm$ 0.03	0.03 $\pm$ 0.03	0.0008 [-0.02, 0.02]	0.95